32. Morphological Operations Using OpenCV – Closing Technique

Aim

To perform **morphological closing** on an image using OpenCV.

Procedure

1. Import the required OpenCV library.
2. Read the input image.
3. Convert the image into grayscale.
4. Create a structuring element (kernel).
5. Apply the **closing operation** using dilation followed by erosion.
6. Display the original and processed images.
7. Save the output image.

Program

```
import cv2

import numpy as np


# Read the input image

img = cv2.imread("input.jpg")


# Convert to grayscale

gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)


# Create structuring element

kernel = np.ones((5, 5), np.uint8)
```

```
# Apply morphological closing
closing = cv2.morphologyEx(gray, cv2.MORPH_CLOSE, kernel)


# Display images
cv2.imshow("Original Image", img)
cv2.imshow("Closing Image", closing)


cv2.imwrite("closing_output.jpg", closing)


cv2.waitKey(0)
cv2.destroyAllWindows()
```

Input

Output

Result

Thus, the **morphological closing operation** was successfully performed using OpenCV. It helps to **fill small holes and close small gaps** in objects while preserving their overall shape.